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OPTIMIZED DIGITAL SUPPORT SERVICE USING LLM AND RETRIEVAL-AUGMENTED GENERATION

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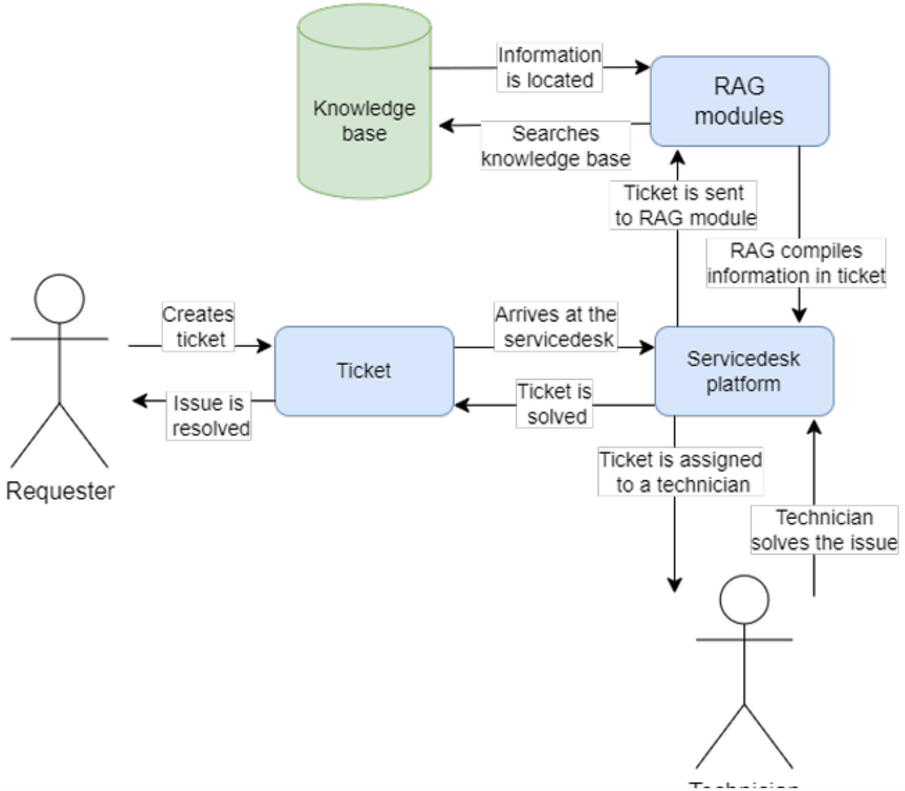
Agenda

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2. LLM & RAG vs Traditional Support Systems
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5. LLM & RAG System Architecture &Design
6. Working Mechanisms
7. Experiment & Results
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Introduction

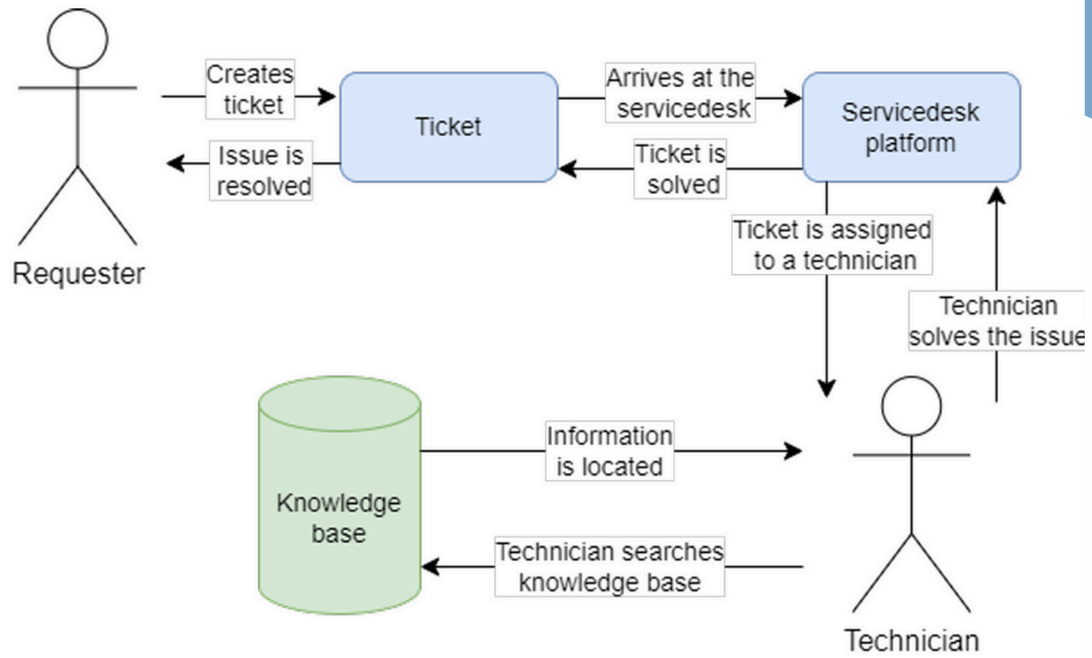
- **LLMs** help IT technicians as digital co-workers for faster issue resolution.
- **RAG** (Retrieval-Augmented Generation) adds domain knowledge for accurate, contextual responses.
- **EBR** (Embedding-based retrieval) uses unstructured data to find relevant solutions.
- Reduces manual effort by summarizing docs and past tickets.
- Study evaluated in Södra’s support system.

LLM & RAG



- RAG module embeds keywords directly into the ticket, reducing technician's resolution time.

Traditional Support Systems



- Technician starts with a raw, unprocessed ticket and must build context from scratch.

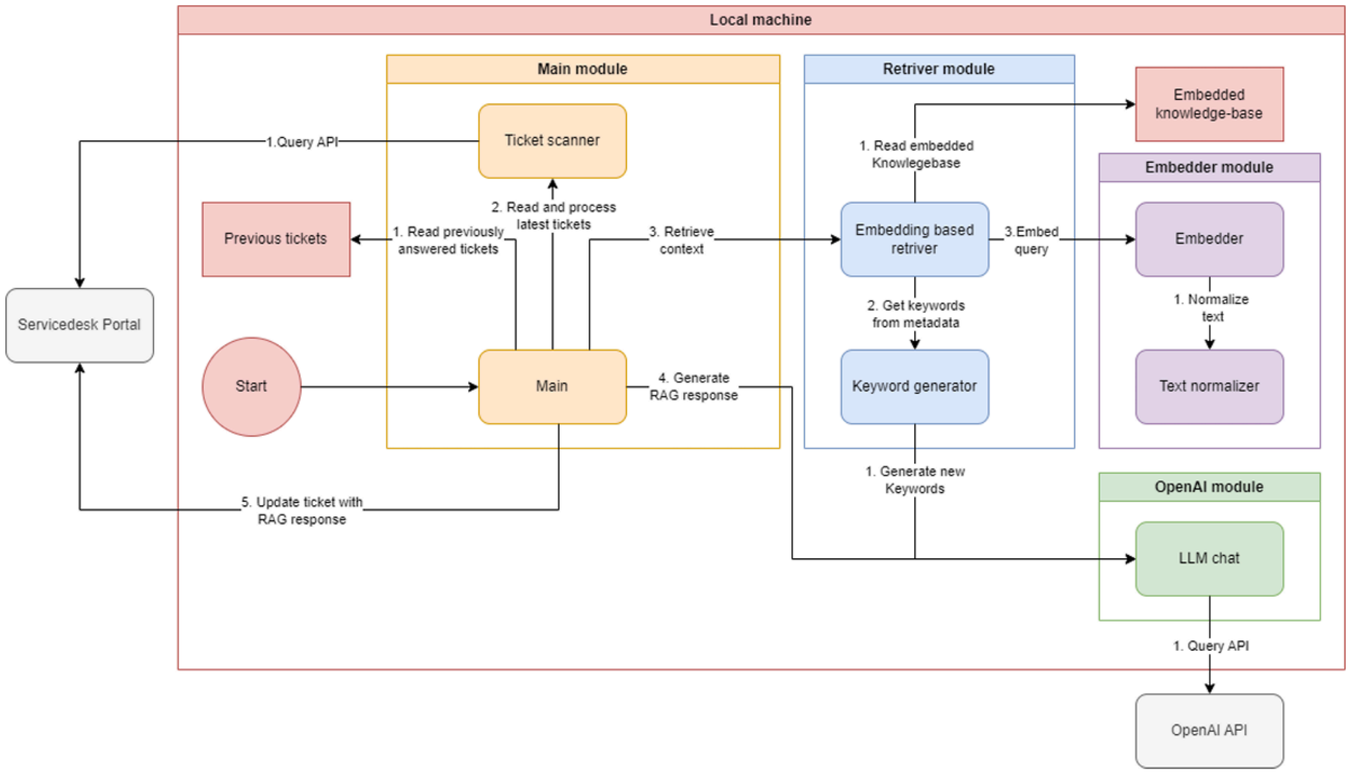
Key findings

- **High Technician Engagement:** Technicians actively interact with the system, removing ~60.6% of inadequate solutions.
- **Low Overall Retention:** Only ~30% of suggested solutions are kept in final responses.
- **Practical Impact:** Solutions may be removed from final responses but still aid technicians internally, as seen in clusters like IT equipment disposal (~61.7% retention).

Advantages

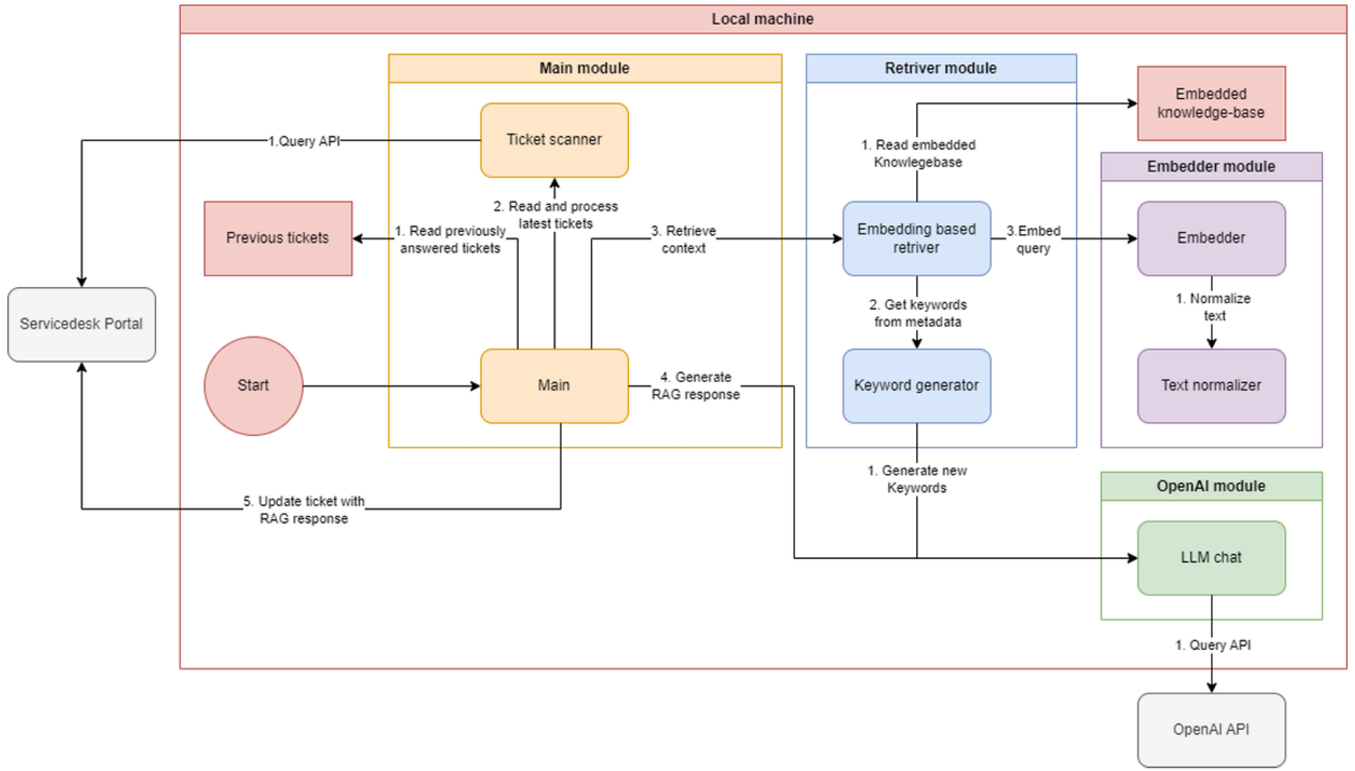
- **Automated workflow solutions:** Pre-generated solutions help technicians start tasks efficiently and reduce mental load.
- **Divergent Feedback:** Solutions are helpful to some, disruptive to others.
- **Domain Expertise is Crucial:** The system’s quality depends on expert data—LLMs need domain-specific training for accurate solutions.

LLM & RAG System Architecture & Design



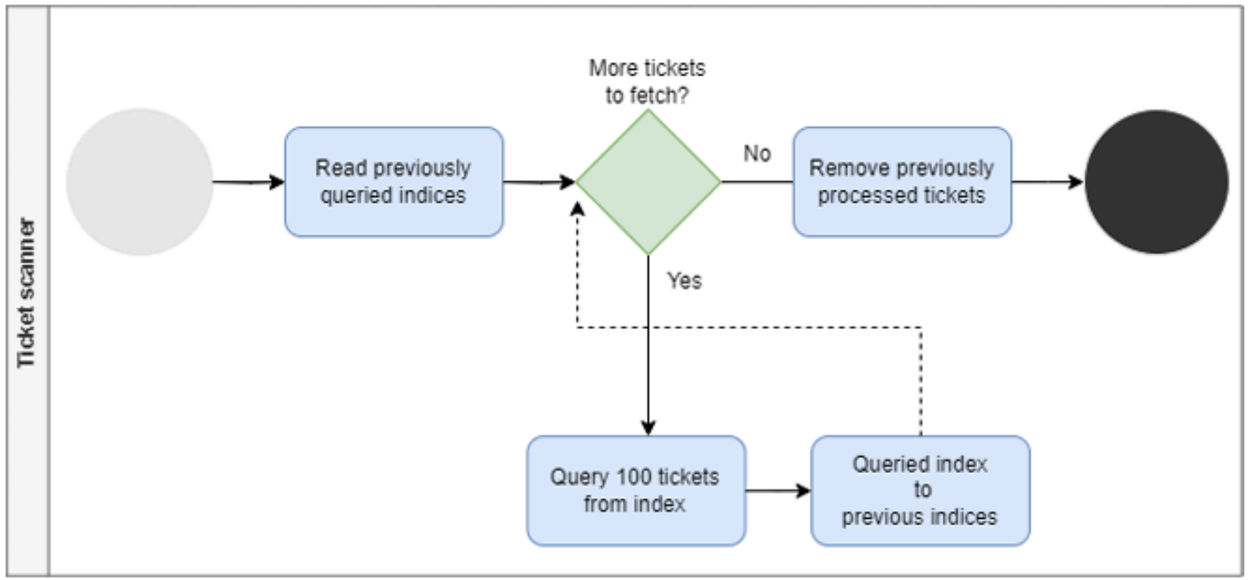
- Process user query from Servicedesk Portal.
- Main Module analyzes new & previous tickets
- Retriever Module uses two methods at the same time .
  1. Vectorizes query to find semantically similar content in the knowledge base.
  2. Extracts keywords from the ticket's metadata for a targeted search.

LLM & RAG System Architecture & Design

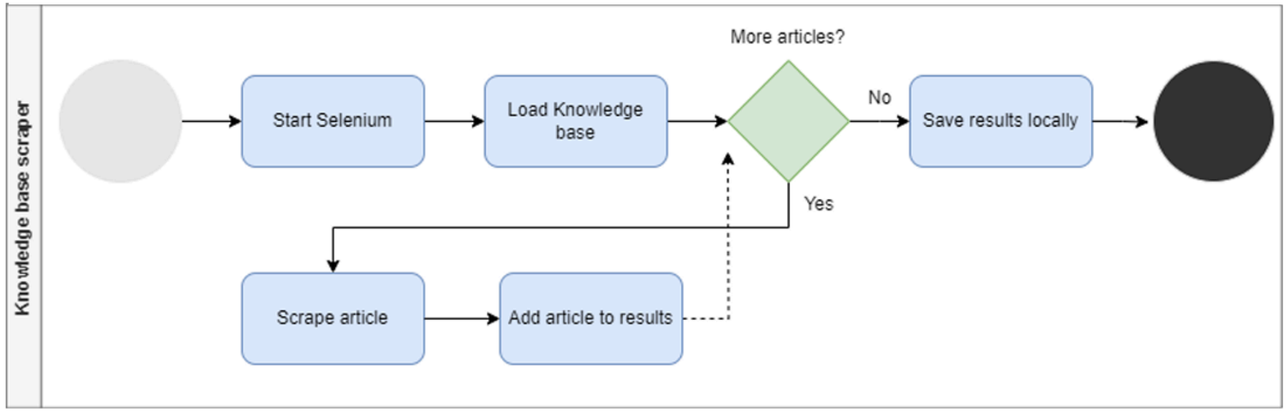


- LLM uses the keywords to generate contextual reponses
- Main Module combies factual data from the knowledge-base with contextual responses from the LLM to generate a RAG response.
- Final RAG response is sent to Servicedesk Portal to update the ticket and resolve the user's query

Working Mechanisms

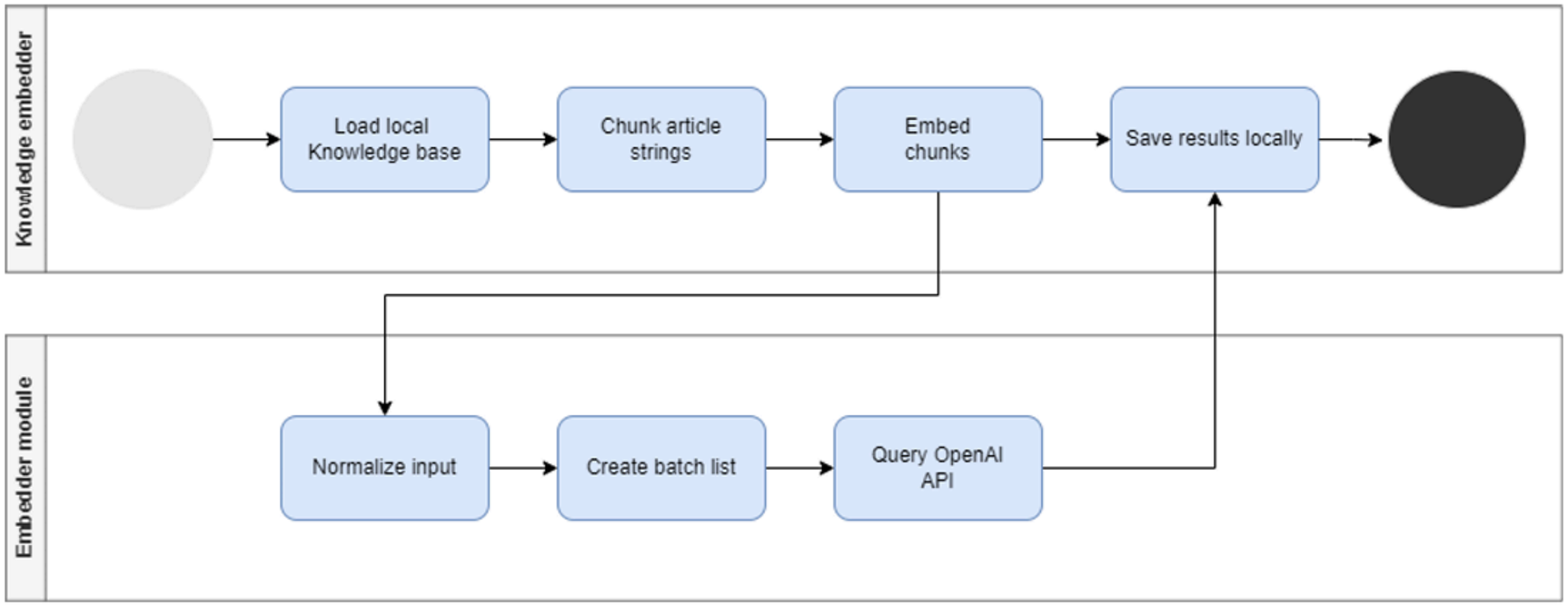


- **Automated Ticket Scanning:** Auto-fetches up to 100 new/open tickets via API.



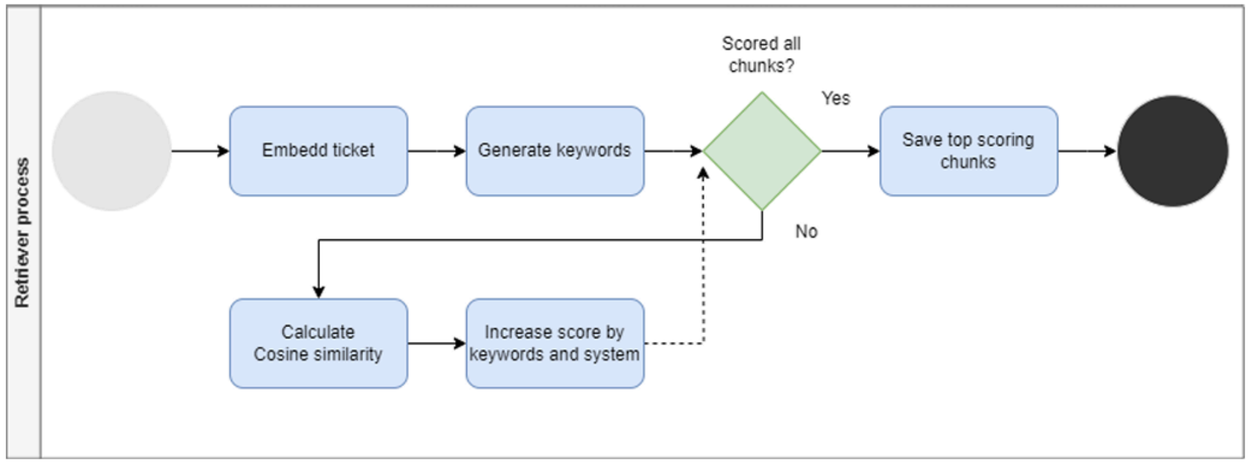
- **Knowledge Base Scraping:** Selenium scraper auto-extracts and stores all SharePoint articles locally.

Working Mechanisms



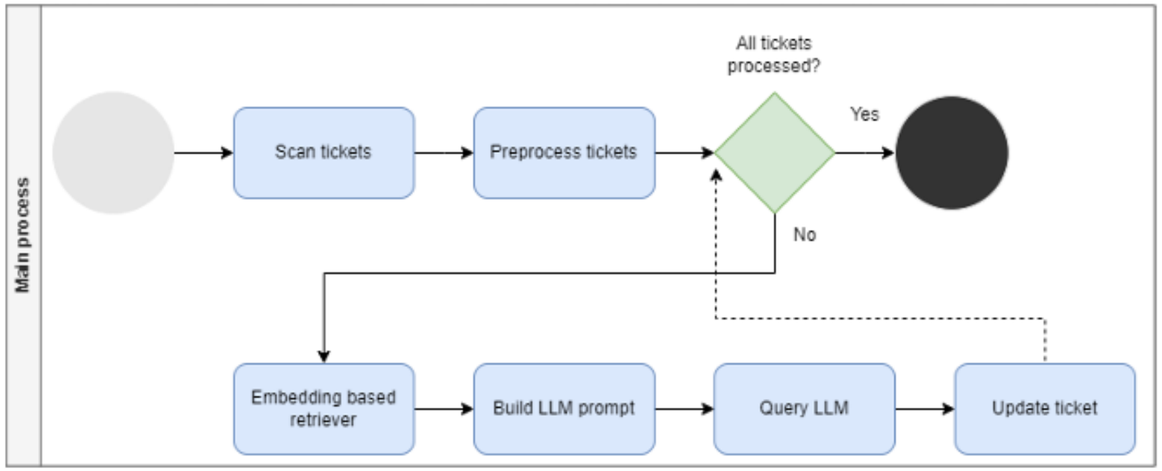
- **Smart Chunking:** Breaks content into chunks and converts them into vector embeddings.
- **Smart DB:** Builds a local vector memory to power contextual retrieval.

Working Mechanisms



- **Ticket Embedding:** Converts ticket content into embeddings for direct knowledge base comparison.
- **Smart Search:** Finds semantically similar knowledge base chunks to the ticket.
- **Hybrid Scoring:** Boosts chunk relevance using dynamic keywords and ticket metadata.
- **Context Filtering:** Extracts the top-scoring chunks for model context.

Working Mechanisms



- **LLM Input Setup:** Structures instructions and ticket data for effective LLM processing.
- **Solution Generation:** LLM produces a response using only the provided ticket and context.
- **Smart Formatting:** Automatically adds disclaimer, solution, and source links to the LLM response.
- **Automated Ticket Closure:** Updates the ticket with the solution and continues processing new tickets.

Experiment & Results

- **Setup:** Two-week live test; 401 of 532 tickets analyzed for real-world performance.
- **Key Result:** 38.4% of AI solutions were adopted; 60.6% removed, 1% modified.
- **Ticket-Type Performance:**Highest success in hardware (100%), followed by VPN (75%) and IT disposal (61.7%).
- **Technician Insight:** Useful for simple tickets (4/5), but low active adoption for complex tasks (2.33/5).

Challenges & Limitations

- **Limited Integration**: LLM used solely in ticket resolution
- **Language Bias:** Non-English tickets may yield less accurate or relevant solutions due to English-centric training.
- **Metrics Missing**: Lack of BLEU/ROUGE/METEOR limits objective assessment and cross-study comparison.
- **Limited Sample:** study size and excluded teams reduce diversity; expanding the sample would strengthen findings.
- **Variable Engagement:** Technician interaction ranged from active to minimal, affecting outcome consistency.
- **Time Impact Unknown:** Efficiency gains weren’t tracked; measuring them could clarify benefits.

Conclusion

- **Primary Goal**:RAG automation eased technician workload and streamlined ticket resolution.
- **Practical importance:** Technicians choosing to retain and use the AI-generated solutions in 38.4% of cases.
- **Key Challenges:** System integration, unstructured data, and Swedish language adaptation.
- **Positive Outcome:** Enhanced resolution process and strong technician involvement.
- **Upcoming Tasks:**Track efficiency gains and expand the study for broader validation.

Future outlook

- **Performance Tracking:** Apply BLEU/ROUGE and time metrics to measure solution accuracy and operational impact.
- **Workflow Expansion:** Automate classification, routing, and follow-up suggestions alongside solution generation.
- **Smarter AI:** Improve accuracy with diverse, multilingual, and context-aware models learning from history and user behavior.
- **Smart Feedback & ROI:** Model adapts from edits while measuring financial and productivity impact.

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THANK YOU

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